



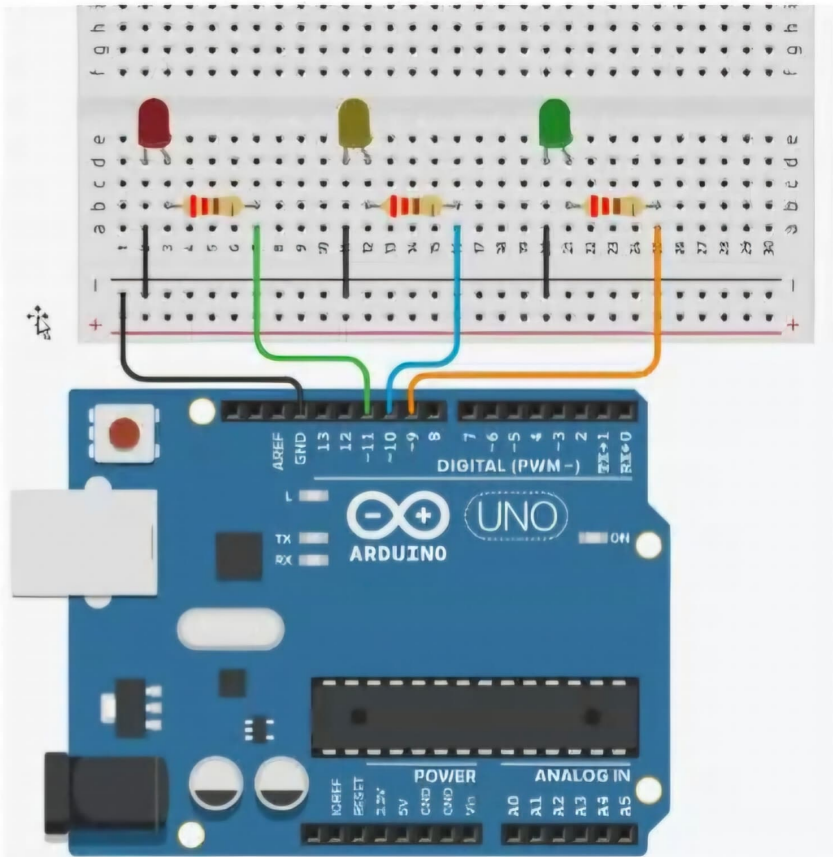
sketch_jul16d.ino

```
1 int redled=12;
2 int time=1000;
3
4 void setup() {
5   // put your setup code here, to run once:
6   pinMode(redled,OUTPUT);
7   pinMode(LED_BUILTIN,OUTPUT);
8 }
9
10 void loop() {
11   // put your main code here, to run repeatedly:
12   digitalWrite(redled,HIGH);
13   digitalWrite(LED_BUILTIN,LOW);
14   delay(1000);
15
16   digitalWrite(redled,LOW);
17   digitalWrite(LED_BUILTIN,HIGH);
18   delay(1000);
19 }
20
```

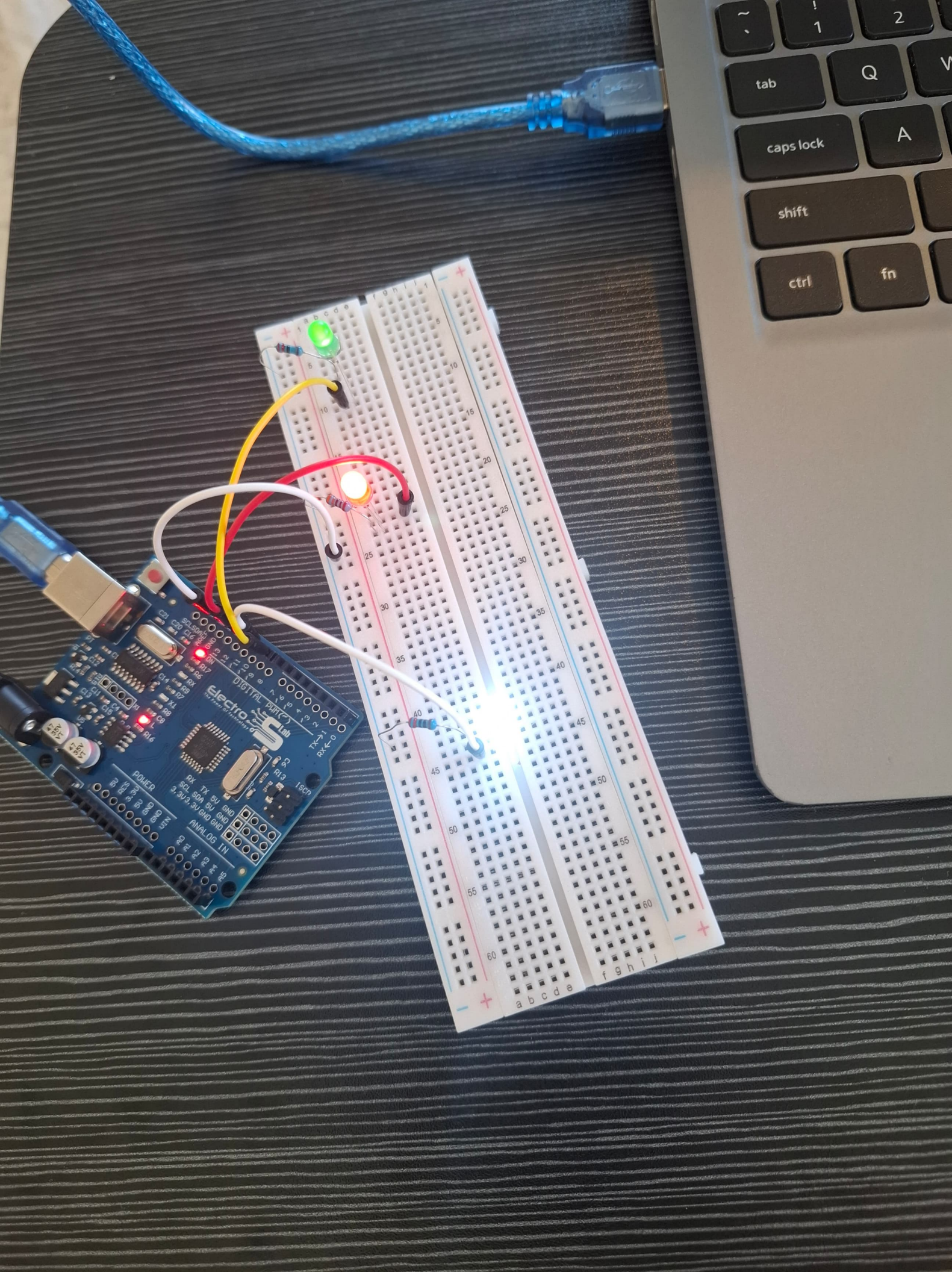
sketch_jul16d.ino

```
1 int redled=12;
2 int time=1000;
3
4 void setup() {
5   // put your setup code here, to run once:
6   pinMode(redled,OUTPUT);
7   pinMode(LED_BUILTIN,OUTPUT);
8 }
9
10 void loop() {
11   // put your main code here, to run repeatedly:
12   digitalWrite(redled,HIGH);
13   digitalWrite(LED_BUILTIN,LOW);
14   delay(time);
15
16   digitalWrite(redled,LOW);
17   digitalWrite(LED_BUILTIN,HIGH);
18   delay(time);
19 }
20
```

Board connection




```
1  int time=1000;
2  int redled=12;
3  int whiteled=9;
4  int greenled=8;
5
6  void setup() {
7      // put your setup code here, to run once:
8      pinMode(redled,OUTPUT);
9      pinMode(whiteled,OUTPUT);
10     pinMode(greenled,OUTPUT);
11 }
12
13 void loop() {
14     // put your main code here, to run repeatedly:
15     digitalWrite(redled,HIGH);
16     digitalWrite(whiteled,HIGH);
17     digitalWrite(greenled,HIGH);
18     delay(time);
19
20     digitalWrite(redled,LOW);
21     digitalWrite(whiteled,LOW);
22     digitalWrite(greenled,LOW);
23     delay(time);
24 }
25
```

RGB LED

The RGB Light is a compound LED that has the main three lights: red, green, and blue.

The RGB consists of three lights, each one has its own pin. We can turn ON the needed color by sending electricity at the right pin:

- 1- We can have one of the main colors
- 2- We can have a white color if we turn ON the three pins
- 3- We can have a different color if we combine different colors together

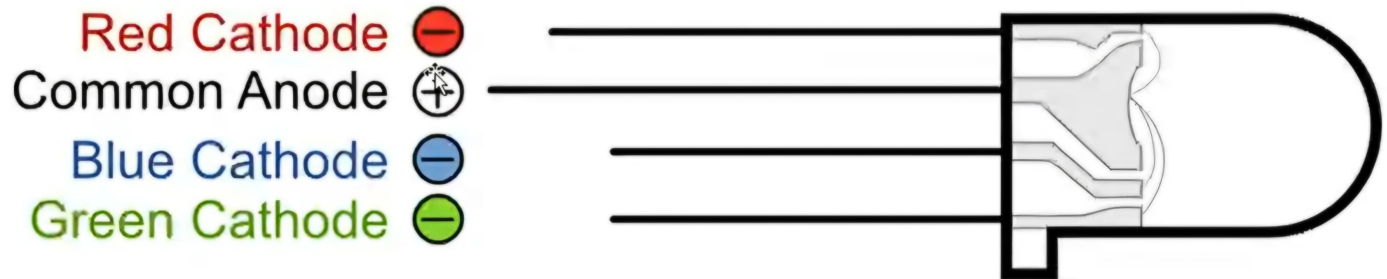


Pin Types for RGB LED

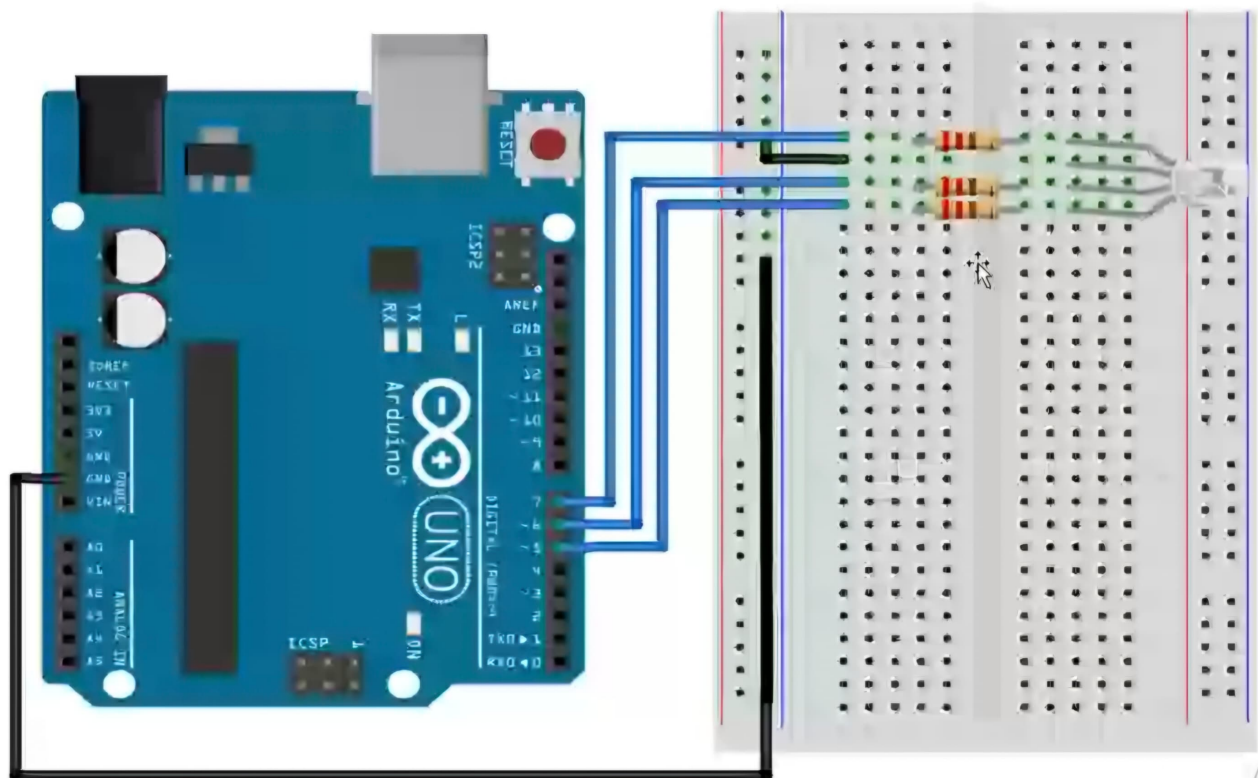
The RGB LED rope has 4 pins:

1. One is for VCC,
2. 3 is for the GND red, blue, green

Depending on the signal frequency, the LEDs will glow and change color.



Board connection



sketch_jul16h.ino

```
33 digitalWrite(red,HIGH); //RED+GREEN=YELLOW
34 digitalWrite(green,HIGH);
35 delay(time);
36 digitalWrite(red,LOW);
37 digitalWrite(green,LOW);
38
39 digitalWrite(green,HIGH); //GREEN+BLUE=LIGHT BLUE
40 digitalWrite(blue,HIGH);
41 delay(time);
42 digitalWrite(green,LOW);
43 digitalWrite(blue,LOW);
44
45 digitalWrite(red,HIGH); //RED+GREEN+BLUE=WHITE
46 digitalWrite(blue,HIGH);
47 digitalWrite(green,HIGH);
48 delay(time);
49 digitalWrite(red,LOW);
50 digitalWrite(blue,LOW);
51 digitalWrite(green,LOW);
52 }
```

```
1  int red=7;
2  int green=6;
3  int blue=5;
4  int time=1000;
5
6  void setup() {
7      // put your setup code here, to run once:
8      pinMode(red,OUTPUT);
9      pinMode(green,OUTPUT);
10     pinMode(blue,OUTPUT);
11 }
12
13 void loop() {
14     // put your main code here, to run repeatedly:
15     digitalWrite(red,HIGH);
16     delay(time);
17     digitalWrite(red,LOW);
18
19     digitalWrite(green,HIGH);
20     delay(time);
21     digitalWrite(green,LOW);
22
23     digitalWrite(blue,HIGH);
24     delay(time);
25     digitalWrite(blue,LOW);
26
27     digitalWrite(red,HIGH);
28     digitalWrite(green,HIGH);
29     delay(time);
30     digitalWrite(red,LOW);
31     digitalWrite(green,LOW);
32     delay(time);
33
```